

S5 System — Industrial IoT Condition Monitoring Awareness Campaign Brief

1. Campaign Target Audience & Operational Context

This campaign targets key operational decision-makers responsible for asset integrity and operational continuity across heavy industrial sectors in Australia.

Target Audience Roles:

- **Asset Managers**
- **Reliability Engineers**
- **Maintenance Superintendents**

Key Industry Sectors:

- **Australian Mining**
- **Heavy Industry**
- **Infrastructure**

Operational Context:

Operations across remote Australian sites face severe geographic and physical infrastructure limitations. The primary operational constraint is the absence of local site power and established communications infrastructure across vast asset footprints.

2. Core Narrative: Calendar-Based vs. Continuous IoT Monitoring

Traditional calendar-based maintenance relies on manual inspection routes that waste skilled labor hours taking periodic snapshot readings. Because inspections occur at fixed intervals, critical mechanical faults frequently develop unnoticed between rounds, leading to unpredicted equipment failure and costly downtime.

The S5 System resolves this operational constraint by deploying ultra-low-power IoT sensors operating on LoRaWAN and NB-IoT protocols directly onto rotating assets. This architecture delivers continuous condition visibility without requiring site power or local communications infrastructure, transforming reactive route data into continuous operational intelligence.

Traditional Calendar-Based Maintenance	S5 Ultra-Low-Power IoT Condition Monitoring
Relies on periodic snapshot data gathered during manual route inspections.	Delivers real-time condition telemetry via LoRaWAN / NB-IoT low-power wireless protocols.
Wastes skilled labor hours conducting routine physical inspections on healthy assets.	Automates continuous asset monitoring, freeing skilled labor to focus exclusively on targeted maintenance.

Misses rapidly developing mechanical faults that occur between scheduled rounds.	Detects emerging mechanical faults continuously before catastrophic asset failure occurs.
Dependent on local site power grid connections and legacy comms infrastructure.	Operates fully autonomously with zero reliance on site power or existing communications infrastructure.

3. Key Business Impact & 3-ROI Case

Transitioning from periodic manual inspections to continuous S5 IoT condition monitoring directly eliminates undetected asset degradation. By providing real-time operational visibility, maintenance teams intervene before failure occurs, unlocking three key financial and operational ROI metrics:

> **Core S5 Value Drivers:**

- > * **30–50%** reduction in unplanned downtime
- > * **20–40%** reduction in maintenance costs
- > * **2–4x** extension in overhaul intervals

4. S5 Product Suite Matrix

The S5 System hardware ecosystem comprises the following dedicated monitoring technologies:

- Vibration Sensor (WearMon / RotaMon)
- LoRa Gateway
- Bolt Tension Sensor
- Wireless Pressure Sensor
- Conveyor Belt Monitor
- Enviro Node

5. Call to Action (CTA) & Campaign Social Metadata

Call to Action:

- > How does your site currently catch mechanical faults before they stop production?

Campaign Hashtags:

#MiningIoT #LoRaWAN #ConditionMonitoring #SmartMining #IndustrialIoT